QCert: Verifying and Extending Quantum Circuit Completeness in Agda and Lean

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A set of circuit equations is complete if every equation between circuits denoting the same linear map follows from it, and minimal if no equation can be derived from the others. Establishing these properties takes long case analyses spanning hundreds of pages, making the details hard to check by hand. To address this, we extend an existing formalization library to QCert in Agda and Lean. In QCert, humans set up a unified framework of circuits, matrix semantics, and group presentations, and state each paper's main theorem in it. LLM coding agents then write the proofs following the papers' arguments, and the proof assistant checks them. A human therefore need only review each paper's relations, gate semantics, and theorem statement. Using QCert, we first check sixteen published results, fifteen on completeness and one on minimality. They cover discrete gate sets such as Clifford, Clifford+T, and Toffoli–Hadamard, on qubits or qudits of every odd prime dimension. We verify fourteen of them directly and correct an error in each of the other two. We then prove a new result, generalizing the completeness of Toffoli–Hadamard circuits from three to any number of qubits. We expect QCert to help with other open completeness problems, such as those for quantum circuits in higher dimensions.

1 Introduction

Rewriting of quantum circuits is a ubiquitous tool in circuit optimization [32, 39, 40, 50], equivalence checking [51], hardware mapping [47], and reasoning about quantum programs [26]. This relies on identifying a set of rewrite rules for a family of quantum circuits that transform a circuit into an equivalent one. Such a theory should be *sound* so that every rewrite preserves the underlying linear map. It should also be *complete* so that any equivalence between circuits can be derived. Additionally, we want it to be *minimal*, with no redundant rules.

Completeness results have been established for a wide range of circuit families, including reversible linear circuits [33, 43], reversible circuits over multiple-control Toffoli gates [27, 28], Clifford [11, 34, 38, 45], CNOT-dihedral [2, 16], two-qubit Clifford+T [13], three-qubit Clifford+CS [14], Toffoli–Hadamard [5], and real Clifford+CH [20] circuits, as well as universal quantum circuits [22, 23]. Completeness has also been obtained for the group of two-level matrices underlying several of these circuit families [12, 19, 26, 30, 35]. Minimal theories have been given for six of these circuit fragments [17] and for all quantum circuits [21].

Completeness and minimality proofs typically require long case distinctions over gates, relations, and normal forms, whose supporting derivations sometimes span hundreds of pages [20, 26, 34]. In particular, relation reductions require intricate circuit manipulations and repeated rewrites, making them prone to human mistakes. For example, circular reasoning can lead us to remove a rule whose derivation still relies on that rule, hidden in a chain of other derivations. Errors can then propagate through dependent derivations, invalidating the entire reduction and making the error source hard to trace by hand.

The length of these proofs and their error-prone nature make it desirable to formally verify their correctness using a proof assistant like Lean or Agda. This would not have been feasible in the past due to the sheer number of lemmas and lengthy rewrites in these papers. Recent LLMs can write correct code for proof assistants, which brings this closer to reality. For example, Anthropic reports that LLMs wrote a 13-million-line Lean proof of Fermat's Last Theorem within 11 days [8]. However, care must still be taken. It is important to manually verify the *alignment* between the machine-verified statement and the intended natural-language result. For instance, [9] showed that a claimed verification of OpenAI's natural-language proof about the Navier–Stokes equations contained mistranslations, so the formal verification was verifying a weaker result by a different proof. A way to solve this is to use a hybrid approach, where humans write and carefully review the definitions and theorem statements [31], coding agents write the proofs, and Agda checks them.

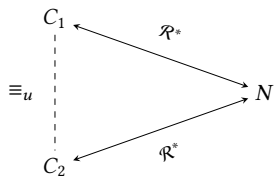
Our contributions. In this work we present QCert, a framework in Agda [18, 41] and Lean [24] for representing and verifying quantum circuit completeness and minimality results, and use it to verify the correctness of sixteen completeness results. This is based on the formalization of qudit Clifford completeness in [11], where the framework [10] was built and the main verification steps were carried out by hand. Since all sixteen completeness results use the same framework of quantum circuits, gates, and matrices over a finitely generated ring, this makes it possible to efficiently check the alignment of the formalized results with the natural-language statements from the original papers. Of these sixteen results, we fully verify fourteen directly, and find fixes to verify the remaining two. For one result, we find a gap in its proof, which we fix, so the original completeness claim still holds. For another result, we find that the theorem is missing one relation, and we show that this relation is independent of the others, so completeness fails without it.

Orientation. In Section 2, we review the two main strategies for proving completeness of a rule set and how minimality is shown. In Section 3, we describe the QCert framework, which adds matrix semantics to the library of circuits and presentations of Bian and Feng [10], together with our formalization workflow and a comparison of the Agda and Lean formalizations. In Section 4, we use QCert to machine-check the completeness and minimality results of sixteen papers. In Section 5, we describe the issues these formalizations found in the source papers. In Section 6, we use QCert to prove a new result, the completeness of Toffoli–Hadamard circuits on any number of qubits. In Section 7, we discuss the current limitations of QCert and our plans to extend it.

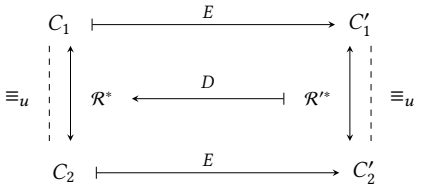
2 Background

A circuit is a word over an alphabet $\mathcal{G}$ of gates. A rewrite rule is then a relation between words over this alphabet, and a rule set $\mathcal{R}$ generates a congruence on circuits. To a circuit C we associate an *interpretation* $[\![C]\!]$, which is its underlying matrix. The completeness results we formalize have matrix entries in the finitely generated rings $\mathbb{Z}\left[1/\sqrt{2}, i\right]$, $\mathbb{Z}\left[1/2, i\right]$, $\mathbb{Z}\left[1/\sqrt{2}\right]$, and $\mathbb{Z}\left[1/2\right]$, which make them exactly representable in Agda. We leave the formalization of universal circuits taking values in $\mathbb{C}$ for future work.

There are two main strategies for proving completeness of a rule set $\mathcal{R}$ over a gate set $\mathcal{G}$, both of which we support. Consider any two circuits C_1 and C_2 over $\mathcal{G}$ that have the same linear map, $[\![C_1]\!] = [\![C_2]\!]$. We need to show that C_1 can be rewritten into C_2 using $\mathcal{R}$. Figure 1a outlines the first strategy. It specifies a unique normal form for each linear map [2, 11, 16, 27, 28, 33, 34, 38, 43, 45], often by giving an exact synthesis algorithm [12, 19, 26, 30, 35]. Completeness follows by proving that every circuit can be transformed into its normal form using the rewrite rules. By uniqueness, C_1 and C_2 have the same normal form N. Reversing the path from C_2 to N and concatenating it with the path from C_1 to N then gives the desired path of rewrites.



(a) The rewrites from C_1 and C_2 to their common normal form N using $\mathcal{R}$ give rewrites from C_1 to C_2 using $\mathcal{R}$.

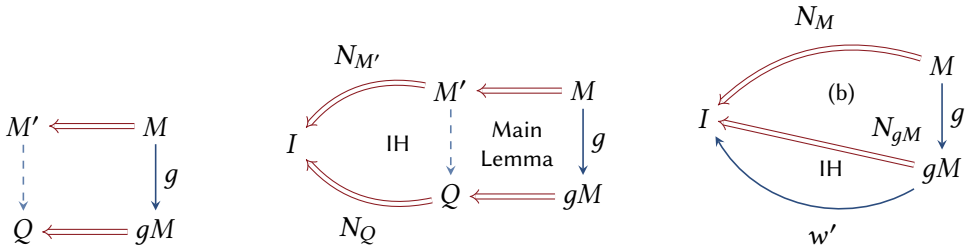


(b) Completeness of $\mathcal{R}'$ gives rewrites between $C_1' = E(C_1)$ and $C_2' = E(C_2)$. Translating this proof back gives rewrites between C_1 and C_2 using $\mathcal{R}$.

Fig. 1. Two strategies for proving completeness of $\mathcal{R}$ over a gate set $\mathcal{G}$. $\overset{\mathcal{R}^*}{\longleftrightarrow}$ means applying a finite sequence of rewrites using $\mathcal{R}$ in either direction, where $*$ denotes zero or more steps.

The exact synthesis results above present groups of matrices rather than circuits (the first block of Table 1). Their generators are *multi-level* matrices. For an $m \times m$ matrix M and distinct indices $a_1, \ldots, a_m$ in $\{0, \ldots, n-1\}$, the *m-level* matrix $M_{[a_1,\ldots,a_m]}$ is the $n \times n$ matrix that acts as M on the standard basis vectors $e_{a_1}, \ldots, e_{a_m}$ and as the identity on the others. For example, $(-1)_{[a]}$ negates e_a, and $\mathrm{X}_{[a,b]}$ exchanges e_a and e_b. Circuits on k qubits have matrices of dimension $n = 2^k$, but these groups are defined in every dimension n. We introduce multi-level matrices for two reasons. The gap we find in Section 5 is in a presentation by one- and two-level matrices [26], and our new result in Section 6 builds on a presentation by one-, two-, and four-level matrices [35].

For these groups, the normal forms come from exact synthesis, and the proof that every circuit can be transformed into its normal form has the same structure for each algorithm. Each matrix, or *state*, has a *level*, built from quantities such as the least denominator exponent of a column, and the algorithm reduces each state M to the identity in steps that lower the level. These steps form the normal word N_M. The *Main Lemma* says that for every state M and generator g, a diagram of relations closes the square formed by g and the algorithm's next steps from M and gM, passing only through states of lower level (Figure 2a). By well-founded induction on the level, the relations then rewrite $N_{gM}\, g$ into N_M (Figure 2b), and by induction on length, every word that reduces M to the identity rewrites into N_M (Figure 2c). Two words with the same matrix therefore rewrite into each other. We formalize these inductions once and prove the Main Lemma for each presentation.



(a) The Main Lemma. (b) Induction on the level. (c) Induction on length.

Fig. 2. Completeness from the Main Lemma. Plain arrows are words, double arrows are steps or runs of the algorithm, and dashed arrows are words through states of lower level.

Figure 1b illustrates the second strategy, which uses an already complete rule set $\mathcal{R}'$ over a different gate set $\mathcal{G}'$ [5, 13–15, 17, 20, 22, 23, 25, 44]. Here, E and D translate circuits from $\mathcal{G}$ to $\mathcal{G}'$ and back, replacing each gate with a circuit over the other gate set. Both E and D preserve circuit composition and the underlying linear map. To transfer completeness from $\mathcal{R}'$ to $\mathcal{R}$, we first show that every rule of $\mathcal{R}'$, translated by D, can be derived using $\mathcal{R}$. We can then replace each step of the proof between C_1' and C_2' by its translated derivation, giving $D(C_1') \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_2')$. It then remains to prove that the rewrite $C_j \overset{\mathcal{R}^*}{\longleftrightarrow} D(C_j')$ is derivable, for which it suffices to prove the round-trip rewrites $g \overset{\mathcal{R}^*}{\longleftrightarrow} D(E(g))$ for each $g \in \mathcal{G}$. Sometimes the two strategies are combined, as in the scaled dyadic extension of [35].

Starting from a complete rule set, we can remove a redundant rule by deriving it from the other rules, thereby preserving completeness. To show that the resulting rule set is minimal, we need to give, for each remaining rule R, an interpretation of circuits that every other rule preserves but R does not. Any derivation using the other rules preserves this interpretation, so R cannot be derived from them [17, 21].

3 The QCert Framework

Our Agda framework QCert states every completeness theorem as a group presentation. Unlike the letters of an ordinary word, gates act on wires. The gates in $\mathcal{G}$ are indexed by the number of wires they act on. Wires are numbered from 0 at the bottom, and a gate always acts on the bottom wires of a circuit. To place it anywhere else, we apply a shift, which takes a circuit on n wires and places it on wires $1, \ldots, n$ of a circuit with $n + 1$ wires. Each relation is stated on as many wires as the paper draws it on. To apply it inside a circuit with more wires, we need *structural rules*, which hold for every circuit, regardless of its gate set. These rules say that an equation can be shifted up by one wire, gates on disjoint wires commute, and a scalar can be placed on any wire. These define the equations of a PROP [36], and are often defined implicitly in completeness papers, but a formalization in Agda must define them explicitly.

We associate to circuits a matrix semantics $[\![-]\!]$ with entries in a finitely generated ring like $\mathbb{Z}\left[1/\sqrt{2}, i\right]$. Hence, $[\![-]\!]$ defines a group homomorphism to a unitary matrix group

G. We build G from the group structures of Agda's standard library, which is checked only once per ring. As matrices, a gate on the bottom wires and a shift are Kronecker products with an identity, so the extension of $[\![-]\!]$ from gates to circuits is defined once for all gate sets. Since the structural rules are universal, their soundness is also proved only once. Then for each paper, it suffices to check the soundness of its own rules. Soundness and completeness say the relations present the image of $[\![-]\!]$, which is the subgroup of G generated by the gates. In Agda, mathematical structures such as groups are defined as records. The record $\mathcal{R}$ IsSubPresentationOf G contains the semantics $[\![-]\!]$, with proofs of soundness and completeness. The record $\mathcal{R}$ IsPresentationOf G adds surjectivity, so the relations present all of G.

Each formalization ends with its main theorem stated as an IsSubPresentationOf or IsPresentationOf record. For example, Statement (1) is the theorem for the real stabilizer circuits of [38], with the relations R1–R16 of its Figure 1 lifted to n wires. Statement (2) is the theorem for the two-level matrices of Hadamard-Π [26], with the relations of its Figure 6.

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presentation : {n : ℕ} → (n VRel,_===_) IsSubPresentationOf U (2 ^ n)    (1)
presentation : {n : ℕ} → (_===_ {n}) IsPresentationOf (O n)              (2)
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Agda checks every proof, so a reader only needs to check that each theorem statement matches its paper's result. Since every theorem is one of these two records, it suffices for a reader to check only four components: the relations transcribed from the paper, the target group, the matrix of each gate, and the line assigning $[\![-]\!]$ to the record. All sixteen statements are collected in a single root module, so the reader can check them in one place.

3.1 Formalization Workflow

Our workflow has three parts. First, before formalizing a paper, we check its rewrite rules, its intermediate results, and each line of its derivations with exact matrices in short scripts. This caught most typos before any Agda code was written. Second, we state each main theorem in QCert before a coding agent starts, so the agent only supplies the proof. Finally, for long derivations, the agent writes a script that turns each line of the paper into one step of the Agda proof, so a failing step points to a specific line of the paper.

The main computational cost came from Agda's conversion checker. If two terms are syntactically identical (i.e., written in the same way), it accepts them immediately. Otherwise, it evaluates both, which can take exponential time for a long matrix product. For such products, we keep circuit interpretations abstract and decide matrix equality by computing a single Boolean using row operations, which is orders of magnitude faster than comparing matrices entry by entry. We record these techniques in the instructions followed by the agents, so later proofs avoid the same slowdowns.

3.2 Compare Lean and Agda Formalisation

The Lean [24] version of QCert formalizes the same sixteen results. We built it because Lean has a larger community than Agda, and because it gives access to Mathlib [48], Lean's library of formalized mathematics. For example, the Lean version states each target group

as a group of complex or real matrices from Mathlib, and decides matrix equations over an exact ring that maps into them.

4 Formalized Completeness and Minimality Proofs

The sixteen formalizations in Table 1 contain about 420,000 lines of Agda code and use no postulates, and each theorem is as general as the result in its source paper. Moreover, all matrices are computed exactly. Their entries lie in algebraic number rings, whose elements are given by finitely many integers, so Agda checks automatically whether two matrices are equal. Over the real or complex numbers, no such automatic check exists, which limits the equality checking of QCert to discrete gate sets, as we discuss in Section 7.

Source	Object	Result	Mod	Time
Groups of m-level matrices				
Greylyn [2014]	$U_n(\mathbb{Z}[1/\sqrt{2}, i])$	presentation, $n \leq 4$	31	42
Bian and Selinger [2021]	$U_n(\mathbb{Z}[1/2, i])$	presentation	37	62
Li et al. [2021]	$O_n(\mathbb{Z}[1/2])$, L_n	presentations	68	79
Fang et al. [2026]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	44	44
Clément [2019]	$O_n(\mathbb{Z}[1/\sqrt{2}])$	presentation	40	64
Circuits				
Selinger [2015]	Clifford	sub-presentation	31	34
Bian, Li et al. [2026]	Clifford, odd prime p	presentation of a model	215	546
Makary et al. [2021]	real stabilizer	sub-presentation	30	35
Amy et al. [2018]	CNOT-dihedral	sub-presentation	30	24
Blake [2026b]	phase-affine, prime d	sub-presentations	56	61
Bian and Selinger [2023a]	two-qubit Clifford+T	sub-presentation in U_4	62	118
Bian and Selinger [2023b]	three-qubit Clifford+CS	sub-presentation in U_8	123	250
Clément [2026]	real Clifford+CH	sub-presentation	346	405
Amy et al. [2024]	three-qubit Toffoli–Hadamard	presentation of L_8	201	102
Li et al. [2025]	qutrit Clifford	presentation	89	282
Minimality				
Blake [2026c]	six fragments	minimal, complete	73	183
Total			1,476	2,332

Table 1. Completeness and minimality results formalized in QCert. A presentation gives generators and relations for a whole group, and a sub-presentation for the subgroup generated by the gate set named in the Object column of the Circuits block. L_n is the group of orthogonal matrices $N/\sqrt{2}^j$, where N is an integer matrix and $j \geq 0$. Mod denotes the number of Agda modules, the units that Agda checks one at a time. Time records their typechecking time in seconds from scratch, measured with Agda 2.8 on an Intel Core i5-14400. Mod and Time consider only each row's own modules, not the shared library or the formalizations it builds on.

Most theorems in Table 1 are proved from scratch in QCert against a matrix semantics, with three exceptions. The two rows of [13, 14] build on the Agda proofs that accompany those papers. These proofs relate the circuit relations to two-level relations by the

Reidemeister–Schreier method [37]. We connect them to our formalizations of the two-level matrix presentations in [12, 30], the first two rows of the table, to obtain sub-presentations of $U_4\left(\mathbb{Z}\left[1/\sqrt{2}, i\right]\right)$ and $U_8\left(\mathbb{Z}\left[1/2, i\right]\right)$. The qudit Clifford theorem of [11] is the only one whose semantics is not a matrix group. Its group presentation lifts a presentation of the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}_p)$ with Pauli corrections and global phases, and its semantics is built in the same way.

5 Corrections to the Source Papers

The formalizations in Table 1 identify two kinds of issues in the source papers. Most are typos, and some of them appear in statements that the proofs rely on. These are easy to fix. Two others are harder to repair. One is a gap in a proof, and the other is a relation missing from a theorem. We describe both below.

The gap is in the proof of Theorem 4.14 of Hadamard-Π [26], which states that the relations in its Figure 6, over the one- and two-level matrices $(-1)_{[a]}$, $\mathrm{X}_{[a,b]}$, and $\mathrm{H}_{[a,b]}$, present the group $O_n\left(\mathbb{Z}\left[1/\sqrt{2}\right]\right)$. One case of the proof relies on Lemmas A.16 and A.17, which have counterexamples in dimension 6, and the case analysis built on them misses a matrix in dimension 9. The authors have confirmed this gap (personal communication). Their main result is however still correct: Clément's unpublished notes [19] present the same group with different relations, and his case analysis covers the missing case, using which the formalization of Theorem 4.14 is completed. We also prove that the two sets of relations are equivalent in every dimension and formalize Clément's proof [6].

The relation is missing from Theorem 5.2 of [5]. It states that the relations in its Figures 4 and 7, over the Toffoli and $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ gates, present the group $O_8\left(\mathbb{Z}\left[1/2\right]\right)$. This is the first of two steps towards presenting the scaled dyadic orthogonal group L_8 by three-qubit Toffoli–Hadamard circuits. The relation r_{K} in its Section 5.5, which splits K on two qubits into two copies controlled by the third qubit (Section 6.2), is in neither figure. Nor can it be derived from them. Hence, Theorem 5.2 fails as stated, and we have informed the authors. Our formalization [7] shows that r_{K} is independent by giving an interpretation of circuits that every other relation preserves but r_{K} does not, and proves the presentations of $O_8\left(\mathbb{Z}\left[1/2\right]\right)$ and L_8 with r_{K} restored.

6 A New Result: Toffoli–Hadamard Completeness for Any Number of Qubits

The Toffoli–Hadamard gate set $\{\mathrm{X}, \mathrm{CX}, \mathrm{CCX}, \mathrm{H}\}$ adds the Hadamard gate to the classical reversible gates X, CX, and CCX. It is computationally universal, meaning that it can efficiently simulate arbitrary quantum computations to arbitrary accuracy [1, 46]. This makes it useful in applications such as Grover-based AES key search [29] and fault-tolerant computation using transversal gates and error correction [42]. Toffoli–Hadamard circuits also have a simple arithmetic structure, which makes them useful for exact synthesis and verification. Their matrices are orthogonal and have a common denominator that is a power of $\sqrt{2}$, a property exploited by exact synthesis algorithms [3, 4]. A complete set of rewrite rules for these circuits would support symbolic proofs of circuit equivalence and provide a basis for verifying circuit optimizations. In this section, we seek a complete theory stated directly as circuit equations, on any number of qubits and without ancillas.

6.1 Preliminaries

Besides Toffoli–Hadamard circuits, we consider the Toffoli–K gate set $\{\mathrm{X}, \mathrm{CX}, \mathrm{CCX}, \mathrm{K}\}$, where $\mathrm{K} = \mathrm{H} \otimes \mathrm{H}$ [4, 5, 35]. Recall that

$$\mathrm{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathrm{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \mathrm{K} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

We number the qubits from 0, with qubit 0 the most significant, so the basis state $|x_0 x_1 x_2\rangle$ of three qubits is the basis vector $e_{4x_0+2x_1+x_2}$. We write X_j and H_j for the X and H gates on qubit j, and $\mathrm{K}_{j,k}$ for $\mathrm{H}_j\mathrm{H}_k$.

The interpretations of these circuits are orthogonal matrices. We write $O_n(\mathbb{Z}[1/2])$ for the group of $n \times n$ orthogonal matrices with entries in the ring of dyadic rationals $\mathbb{Z}[1/2]$. We write L_n for the group of $n \times n$ orthogonal matrices of the form $N/\sqrt{2}^j$, where N is an integer matrix and $j \geq 0$. The first group is a subgroup of the second, of index 2 when n is even [35]. On n qubits, the interpretations of Toffoli–K circuits lie in $O_{2^n}(\mathbb{Z}[1/2])$, and those of Toffoli–Hadamard circuits lie in L_{2^n}. With ancillas, every matrix in these groups is the interpretation of some circuit [4]. Without ancillas, as in this section, the circuits may generate only a subgroup. Hence, we state completeness on n qubits as a sub-presentation (see Table 1).

Li et al. [35] gave presentations of both groups in every dimension n. Their generators are multi-level matrices (Section 2) rather than gates. The generators of $O_n(\mathbb{Z}[1/2])$ are the one-, two-, and four-level matrices

$$(-1)_{[a]}, \qquad \mathrm{X}_{[a,b]}, \qquad \mathrm{K}_{[a,b,c,d]}, \qquad 0 \leq a < b < c < d < n,$$

and a finite list of relation schemas, stated independently of n, presents this group [35, Theorem 4.10]. For even n, adding the generator $\mathrm{I} \otimes \mathrm{H} = \mathrm{diag}(\mathrm{H}, \ldots, \mathrm{H})$ and four relations gives a presentation of L_n [35, Theorem 5.7]. A few generators are gates. On three qubits, for example, the Toffoli gate controlled by qubits 0 and 1 is $\mathrm{X}_{[6,7]}$, and $\mathrm{I} \otimes \mathrm{H}$ is H_2. Most generators, however, are not gates of either gate set. For instance, $\mathrm{K}_{[4,5,6,7]}$ applies $\mathrm{K}_{1,2}$ only when qubit 0 is in the state $|1\rangle$. So these presentations are not yet equational theories for circuits.

6.2 Previous Completeness Proofs

In 2021, Li et al. [35] found a complete set of rewrite rules for the group $O_n(\mathbb{Z}[1/2])$, whose generators are one-, two-, and four-level matrices. These generators are matrices rather than gates, so the rules did not yet apply to circuits. In 2023, Vilmart [49] gave a complete set of rewrite rules for Toffoli–Hadamard computations on any number of qubits, in the sum-over-paths formalism. Its terms, however, represent general linear maps rather than circuits. Completeness for such a class of linear maps does not in general yield complete rewrite rules for circuits, because a derivation between two circuits may pass through terms that are not circuits. In 2024, Amy et al. [5] used the previously known fact that on three qubits, the gates X, CX, CCX, and $\mathrm{H} \otimes \mathrm{H}$ generate a finite group, the automorphism group of the E_8 lattice. They first turned its well-known Coxeter presentation into rewrite

rules for these circuits. They then used the rules of Li et al. [35] for $n = 8$ to add the gates CCZ and H. This gave a complete set of rewrite rules for three-qubit Toffoli–Hadamard circuits, once a missing relation is restored (Section 5). Their method stops at three qubits. On four or more qubits, the same gates generate an infinite group, so there is no finite group to start from.

Figure 3 details the proof in three steps. *Step 1* (Figure 3a): translate Toffoli–K circuits U_j with equal matrices into reflection sequences $F_j = E_0(U_j)$. The known complete rules $\mathcal{R}_{E_8}$ connect F_1 and F_2. Prove that each reflection rule follows from circuit rules $\mathcal{R}_0$, and that translating back recovers the original circuits. Thus D_0 turns the reflection proof into a circuit proof, establishing completeness of $\mathcal{R}_0$. *Step 2* (Figure 3b): add CCZ and a helper that applies K to qubits 1 and 2 when qubit 0 is 0. These circuits B_j represent $O_8(\mathbb{D})$, where $\mathbb{D} = \mathbb{Z}[1/2]$. Translate them into matrix-operation sequences $M_j = E_1(B_j)$. Derive Li et al.'s complete rules $\mathcal{R}_{\mathrm{Li}}$ [35] from $\mathcal{R}_0$ and additional circuit rules, including the restored relation

$$r_{\mathrm{K}} : \quad \mathrm{K}_{1,2} = \mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0\,\mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0.$$

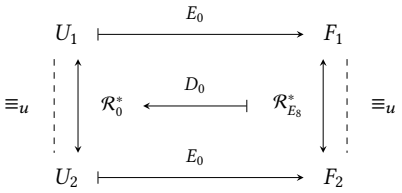
Here $\mathrm{K}_{[4,5,6,7]}$ applies $\mathrm{K}_{1,2}$ when qubit 0 is $|1\rangle$, and the helper $\mathrm{X}_0\,\mathrm{K}_{[4,5,6,7]}\,\mathrm{X}_0$ applies it when qubit 0 is $|0\rangle$, so together they apply $\mathrm{K}_{1,2}$. Translating proofs back by D_1 gives completeness of the resulting rules $\mathcal{R}_1$ [5, Theorem 5.2, with r_{K} restored]. *Step 3* (Figure 3c): extend $\mathcal{R}_1$ to $\mathcal{R}_2$ with Hadamard rules. These move Hadamards to the end, changing intervening gates, and cancel pairs, giving $C_j \overset{\mathcal{R}_2^*}{\longleftrightarrow} B_j H_2^{\epsilon}$, where H_2 acts on qubit 2 and $\epsilon \in \{0, 1\}$. Equal matrices force the same ϵ: B_j has rational entries, whereas $B_j H_2$ has an irrational entry. Cancel the common final factor, connect B_1 and B_2 using $\mathcal{R}_1$, and restore the factor. Hence $\mathcal{R}_2$ is complete [5, Theorem 6.3].

Recently, Clément [20] found a complete set of rewrite rules on any number of qubits for a closely related family, the real Clifford+CH circuits over the gates H, Z, CZ, and the controlled Hadamard gate CH. His rules consist of eighteen equations on at most three qubits and one equation schema, which gives an equation on n qubits for each $n \geq 5$. The rules we find for Toffoli–Hadamard circuits in Section 6.3 have the same shape, a few equations on a small number of qubits together with one schema over the number of qubits. Our proof also follows his. He translates circuits back and forth to a presentation by one- and two-level matrices, as in Figure 1b, and we do the same with the presentations of Li et al. [35].

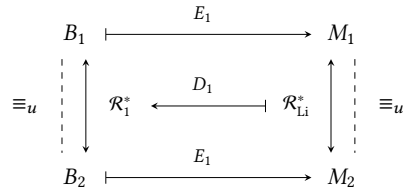
6.3 Completeness for Any Number of Qubits

7 Discussion and Future Work

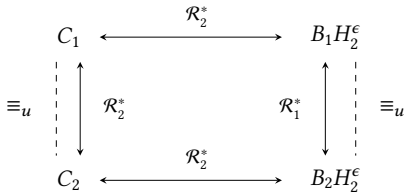
We formalize the results of sixteen papers in QCert, using consistent circuit syntax, a unified definition of group presentations, and shared libraries for rings and their matrix groups. Since all formalizations use these definitions, presentations of the same group from different papers can be compared and proved equivalent. The same libraries, together with the induction that turns an exact synthesis algorithm into a completeness proof, can be reused for new algorithms. Only the case analysis over matrices and generators is specific to each algorithm. As the most error-prone part of the proof, it is where machine checking in QCert matters most.



(a) U_j: Toffoli–K circuits; F_j: E_8 reflection sequences.



(b) B_j: circuits for $O_8(\mathbb{D})$; M_j: matrix-operation sequences.



(c) C_j: Toffoli–Hadamard circuits; $B_jH_2^\epsilon$: at most one final Hadamard.

Fig. 3. Three stages of completeness for three-qubit Toffoli–Hadamard circuits [5], with r_{K} restored (Section 5). Vertices label circuits or operator sequences. Dashed $\equiv_u$ lines assert equal matrices; $\mathcal{R}^*$ arrows denote zero or more rewrites in either direction. E_i translates descriptions, and D_i translates proofs back into circuit rewrites.

At the moment, not every completeness result fits into QCert, as its scope is limited in two ways. First, it supports only discrete gate sets, so it does not cover completeness results for circuits with continuous parameters, such as rotations by arbitrary angles [21–23]. Second, control is not a circuit constructor in QCert, so it does not cover results that treat control as a primitive operation on circuits [15, 25, 44].

We plan to extend the infrastructure of QCert to more families of quantum circuits, including the two above. Completeness is still open for many discrete gate sets in higher-dimensional quantum systems, and their case analyses are often too large to check by hand. In particular, we plan to study the completeness of Clifford circuits in all dimensions with the help of QCert. Beyond completeness, the exact synthesis libraries can be reused for T-count optimization of multi-qubit Clifford+T circuits.

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